Why Does BSE Close Higher? Stale Prices, Thin Trading and a Closing-Auction Experiment

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Data and code: github.com/udit20112007/WORLD-CLOSE

Abstract

In earlier work I found that thousands of Indian companies end each day with two slightly different official closing prices, one on NSE and one on BSE, and that for the least-traded companies BSE's close tends to sit above NSE's. This paper asks why. Using 491,644 pairs of closing prices for 2,212 companies (2025-10-20 to 2026-10-26), I test four explanations. First, BSE's close is often stale for thin stocks: for the least-traded fifth, each 1% rise in a stock's NSE price goes with a gap 6.6 basis points lower, as if BSE had not caught up. Staleness does not explain the bias itself, however: on days when NSE barely moved, BSE still closed 12.0 basis points higher on average. The bias is concentrated on days with almost no trading on BSE. After a gap, BSE does about 84% of the correction the next day, so BSE's close is the noisy price. Thin stocks also trade on BSE mostly on days when they fall on NSE, and a BSE close that does not fully follow the fall sits above NSE's. I date a rise in the typical gap to 2026-01-19 that is not explained by changes in the data, the mix of companies or market volatility. Finally, I use the introduction of a closing auction for stocks with derivatives on 2026-08-03 as a natural experiment. For the most-traded fifth of companies, the typical gap between the two exchanges' closes rose from 3.7 to 8.0 basis points, while other companies barely changed. Separate auctions on each exchange appear to make the two official closes agree less, not more.

1. Introduction

Most large and mid-sized Indian companies are listed on both the National Stock Exchange (NSE) and BSE, and each exchange publishes its own official closing price every day. In an earlier paper (Goyal, 2026; doi:10.5281/zenodo.23666692) I showed that the two closes usually differ by about a tenth of a percent, that the gaps are largest for the least-traded companies, and that they mostly disappear by the next day. One pattern did not fit: for thinly traded companies, BSE tends to close slightly above NSE, and the tendency is too systematic to be chance.

This paper tries to explain that pattern. The earlier paper suggested three candidates, and I add a fourth. The first is that BSE's close is *stale*: if a thin stock does not trade on BSE late in the day, its BSE close may come from an earlier trade and miss the day's later move. The second is the way the close is *formed* when trading is very thin, for example a handful of buyer-initiated trades executing near the ask. The third is *selection*: a stock only appears in the matched sample on days it trades on BSE, and those days may be unusual. The fourth asks which close is closer to the truth, by looking at which exchange moves to close the gap the next day, an idea related to the price-discovery literature (Hasbrouck, 1995).

Stale prices are a well-known problem in finance. When some prices are recorded earlier than others, measured returns become correlated in ways that do not reflect real economic links (Scholes and Williams, 1977; Lo and MacKinlay, 1990). Dual-listed Indian stocks are a clean place to measure this, because the same security has two closing prices on the same

day, and one of them is usually based on far fewer trades.

2. Data and method

I use the same data and sample as the earlier paper: the official daily “bhavcopy” files that both exchanges publish, collected by an automated pipeline, with companies matched across exchanges by ISIN. I keep NSE’s regular equity series, drop shares priced below Rs 20, and require trading on both exchanges that day. The sample has 491,644 stock-days for 2,212 companies from 2025-10-20 to 2026-10-26. In addition to the closing price, this paper uses each exchange’s open, high, low, previous close and number of trades. As in the earlier paper’s company-level analysis, I keep companies with at least 60 matched days, and group them into fifths by their median daily value traded on NSE.

The gap is the BSE close minus the NSE close, divided by the average of the two, in basis points (hundredths of a percent). A positive gap means BSE closed higher. Returns are computed from each exchange’s own previous close, and daily moves larger than 25% are excluded because they usually reflect unadjusted corporate actions. All regressions report standard errors clustered by company.

According to the exchanges’ published methodologies, both NSE and BSE compute the official close as the volume-weighted average price of trades in the last 30 minutes of continuous trading. From 3 August 2026, stocks with derivatives contracts moved to a Closing Auction Session on both exchanges, in which the close is the price of a separate auction held after continuous trading ends (NSE, 2026). Section 8 uses this change as a natural experiment.

3. Are BSE closes stale?

If BSE’s close is stale for a thin stock, it reflects the price from earlier in the day. On a day when the stock rises on NSE, BSE’s close will then lag below NSE’s, and on a day it falls, BSE’s close will stay above. So the gap should move *opposite* to the day’s NSE return. Table 1 regresses the gap on the stock’s NSE return for each liquidity group. For the least-traded fifth, the slope is −6.6 basis points per 1% move (t = −18.7): on a day the stock rises 2% on NSE, BSE’s close is on average about 13 basis points lower than NSE’s. The slope shrinks steadily with liquidity, to −0.4 for the most-traded fifth. Figure 1 shows the pattern: for thin stocks, the gap falls almost in a straight line as the day’s return rises. This is what stale prices predict.

Table 1. The gap against the day’s NSE return. Slope from a regression of the gap (bps) on the stock’s NSE return (%), standard errors clustered by company. Up and down days are moves larger than 0.5%; flat days are within ±0.25%.

Liquidity group	Slope (bps per 1% NSE move)	t	Mean gap, NSE up / down days (bps)	Mean gap, flat days (bps)	BSE higher, flat days
Least-traded fifth	−6.6	−18.7	−8.9 / 38.4	12.0	53.0%
2nd fifth	−1.7	−15.1	−4.3 / 7.3	1.0	49.0%
3rd fifth	−0.3	−8.8	−0.7 / 2.2	−0.2	48.3%
4th fifth	−0.1	−4.8	−0.4 / 1.0	−0.6	45.3%
Most-traded fifth	−0.4	−11.9	−2.1 / 0.6	−1.4	41.9%

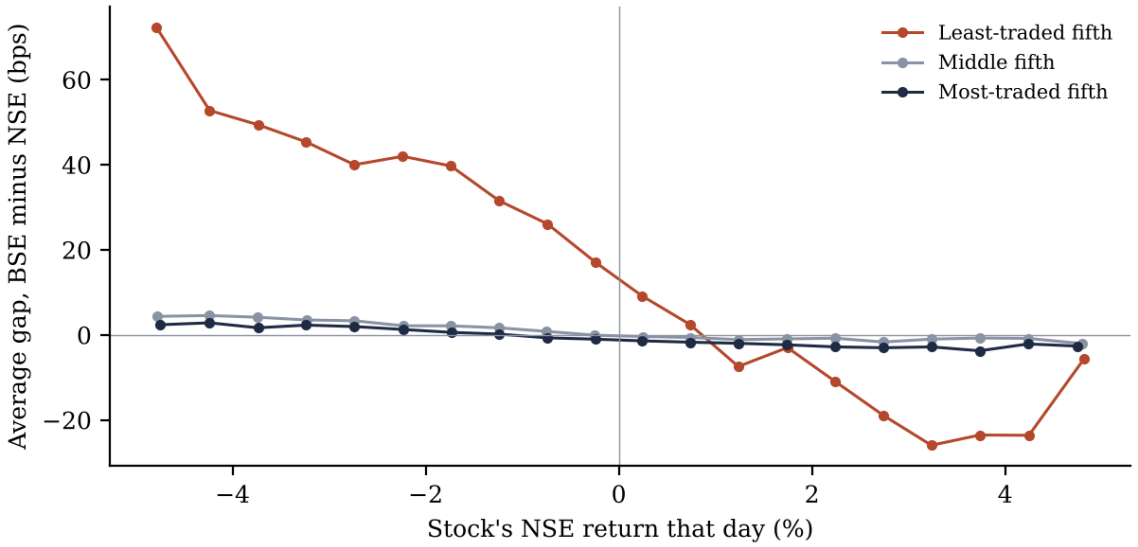


Figure 1. Average gap (BSE minus NSE, bps) against the stock's NSE return that day, in bins of 0.5 percentage points, for three liquidity groups. Bins with fewer than 50 observations are omitted.

Staleness explains why the gap moves with the day's return, but it does not by itself explain why the *average* gap is positive. A stale close only produces a positive average if prices fell on average. The least-traded fifth had an average NSE return of −0.031% a day, and multiplying this by the slope accounts for about 1% of the group's average gap of 16.3 basis points. The more direct check is to look at days when nothing happened. On days when the stock moved by less than 0.25% on NSE, BSE still closed 12.0 basis points higher on average, and was higher on 53.0% of those days. Something other than staleness is pushing BSE's close up. Grouping days by the market as a whole gives the same picture: for the least-traded fifth the average gap was 13.4 basis points on days the median stock rose and 18.5 on days it fell. The response is also lopsided. On days the stock fell by more than 0.5% on NSE, BSE closed 38.4 basis points higher on average, but on days it rose by more than 0.5%, BSE closed only 8.9 basis points lower. Pure staleness would give roughly mirror-image numbers, so this asymmetry is another sign of a separate upward push in BSE's close.

4. Is the bias concentrated on thin BSE days?

If the bias comes from how BSE's close is formed when there is almost no trading, it should be largest on days when BSE trading is thinnest. Table 2 compares such days with other days for the least-traded fifth, using 4 different ways of identifying a thin BSE day. The bias is clearly larger on thin days: on days with 3 or fewer trades on BSE, the mean gap is 37.4 basis points, against 14.1 on other days. This points to the closing-price formation on BSE when only a few trades occur.

Table 2. Thin BSE days, least-traded fifth of companies. The difference is from a regression of the gap on an indicator for a thin day, with standard errors clustered by company.

Measure of a thin BSE day	Share of days	Mean gap, thin / other days (bps)	BSE higher, thin / other	Difference (t)
3 or fewer trades on BSE	9.4%	37.4 / 14.1	56.0% / 54.1%	23.3 (4.8)
BSE high equals BSE low (all trades at one price)	5.6%	39.6 / 14.9	56.4% / 54.1%	24.7 (4.0)
BSE close equals BSE open	7.1%	34.9 / 14.9	56.6% / 54.1%	20.0 (4.0)
BSE volume in the stock's own lowest fifth of days	20.4%	26.9 / 13.6	56.8% / 53.6%	13.3 (6.6)

5. Which close is right?

When the two closes disagree, at least one of them is wrong, and the next day's prices show which. If BSE's close was noise, BSE's price should move back toward NSE's the next day while NSE's barely reacts. If BSE's close contained information NSE had not yet priced, NSE should move toward BSE. I regress each exchange's next-day return on today's gap; Table 3 reports how much of the gap each exchange's move closes. For the least-traded fifth, BSE's move closes 72% of the gap and NSE's only 14%, so BSE does about 84% of the correction. NSE's close is the better price; BSE's close for thin stocks is mostly noise around it. Across all companies, 87% of a gap is closed by the next day, consistent with the earlier paper's finding that gaps do not persist. Figure 2 shows the split by liquidity group. For the most-traded fifth the correction is split almost evenly (50% by BSE), as one would expect when both closes are equally informative.

Table 3. Next-day adjustment. Coefficients from regressions of each exchange's next-day return on today's gap, expressed as the percentage of the gap closed (t-statistics in brackets, clustered by company).

Liquidity group	NSE moves toward BSE (% of gap)	BSE moves back toward NSE (% of gap)	Gap closed next day	Share of correction by BSE
All companies	14.8 (17.0)	72.2 (−41.8)	87%	83%
Least-traded fifth	14.1 (14.2)	71.8 (−34.8)	86%	84%
2nd fifth	16.3 (7.4)	78.8 (−34.9)	95%	83%
3rd fifth	19.9 (4.5)	77.3 (−16.8)	97%	80%
4th fifth	13.2 (2.2)	84.9 (−13.5)	98%	86%
Most-traded fifth	49.7 (12.9)	49.0 (−12.7)	99%	50%

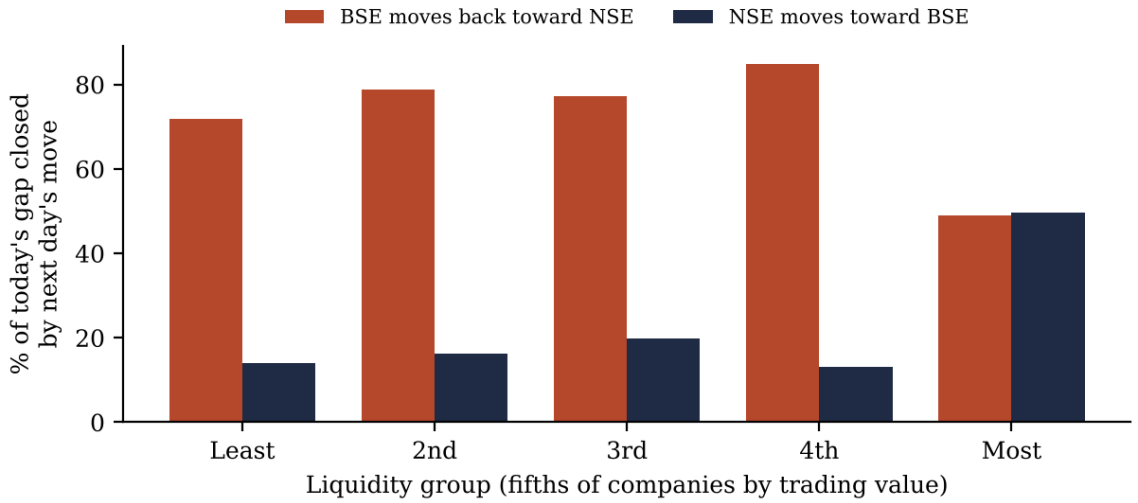


Figure 2. Percentage of today's gap closed by each exchange's move the next day, by liquidity group.

6. Are BSE trading days unusual?

A thin stock only enters the matched sample on days it trades on BSE. The least-traded fifth traded on BSE on 93% of its NSE trading days. If BSE trades happen mostly when there is buying interest, the matched days would be days of rising prices, and the gap measured on those days could be biased. Table 4 compares each company's NSE behaviour on days it also traded on BSE with days it did not. The result goes the other way from buying pressure. For the least-traded fifth, NSE returns are 0.109 percentage points *lower* on days the stock also traded on BSE (t = −3.5). On BSE trading days the stock's NSE close sits at 0.41 of the day's range, against 0.46 on other days (0 is the day's low, 1 the high), so the stock tends to close nearer its low on NSE on those days. Days without any BSE trading are quiet days: NSE value traded is 0.61 times the company's usual level, against 1.03 on days it also traded on BSE. Together with Section 3, this suggests a simple mechanism behind part of the bias. Thin stocks tend to trade on BSE on days when they fall; BSE's close does not fully follow the fall; and so BSE ends those days above NSE.

Table 4. NSE behaviour on days a company also traded on BSE. The within-company difference compares each company with itself, with standard errors clustered by company. Value traded is relative to the company's median day.

Liquidity group	NSE days also traded on BSE	NSE return, BSE days / other (%)	Within-company difference (t)	NSE value vs usual, BSE days / other
Least-traded fifth	92.7%	−0.030 / 0.054	−0.109 (−3.5)	1.03 / 0.61
2nd fifth	99.3%	−0.016 / 0.153	−0.170 (−2.0)	1.00 / 0.67
3rd fifth	99.5%	−0.027 / 0.093	−0.101 (−1.1)	1.00 / 1.00
4th fifth	99.8%	0.017 / 0.104	0.105 (2.5)	1.00 / 0.91
Most-traded fifth	99.9%	0.035 / 0.420	−0.473 (−1.5)	1.00 / 1.75

7. The level shift

The earlier paper's daily series showed the typical gap moving to a higher level during the sample. To date the change, I fit a model in which the daily median gap has one mean before an unknown date and another after, and choose the date that fits best (Andrews, 1993). The best break is 2026-01-19, with a sup-F statistic of 68.0 (the 5% critical value for this kind of test is roughly 9). The median gap rose from 9.6 to 11.4 basis points. The rise appears in every liquidity group. It is not a change in the data: the number of matched companies was 1,933 a day in the ten days before the break and 1,956 in the ten days after. Nor is it a change in the mix of companies: for companies present on both sides of the break, the median gap rose from 9.6 to 10.9 basis points. The market did become more volatile, with the median stock's absolute daily return rising from 0.52% to 0.84%, and the earlier paper found that gaps widen on volatile days. But controlling for the size of each day's market move barely changes the estimated shift, from 1.85 to 1.66 basis points (t = 5.8). The break date coincides with NSE's circular introducing the Closing Auction Session (NSE, 2026), but the auction itself only began in August, so the announcement alone is unlikely to have changed how closes were formed. I cannot explain the January shift with this data.

Table 5. Daily averages before and after the estimated break on 2026-01-19.

Measure (daily average)	Before	After
Median absolute gap across stocks (bps)	9.6	11.4
Companies traded on both exchanges	1,888	1,992
Median share price (Rs)	302	286
BSE share of value traded (%)	6.7	7.1
BSE days with a single price (%)	1.1	1.1

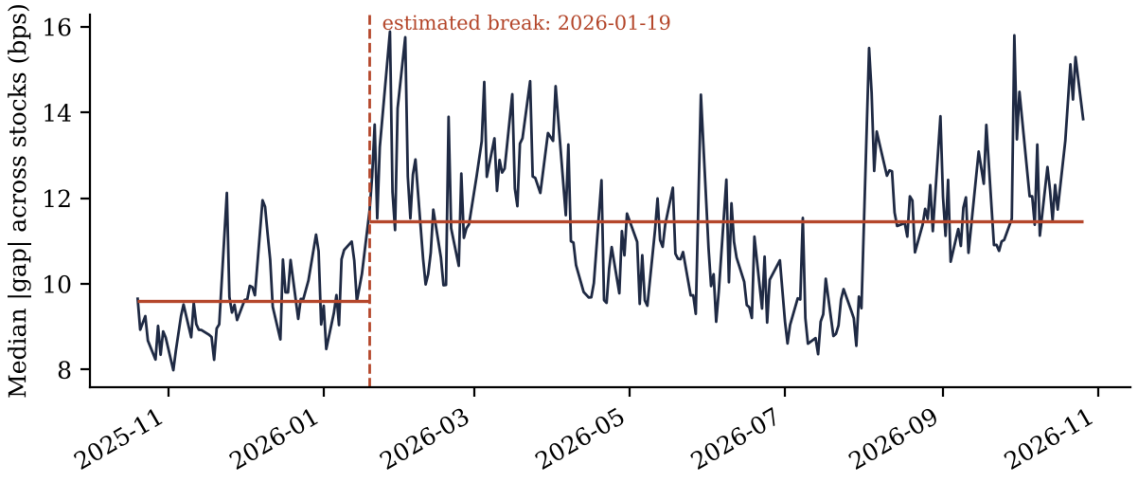


Figure 3. Median absolute gap across stocks each day, with the estimated break date (dashed) and the average level before and after.

8. A natural experiment: the closing auction

On 2026-08-03 NSE and BSE replaced the 30-minute average close with a Closing Auction Session for stocks that have derivatives contracts, while all other stocks kept the old rule (NSE, 2026). This gives a natural experiment: if the way the close is formed matters for the gap, the gap should change for the stocks that moved to the auction and not for the others. I do not have the official list of stocks with derivatives for this period, so I use the most-traded fifth of companies as the treated group. Derivatives are only available on large, heavily traded stocks, so most of them should be in this group. The group also contains companies without derivatives, which kept the old rule; including them can only dilute the measured effect, not create it. I compare the 40 trading days before the change with the 58 days after.

The effect is large and immediate (Figure 4 and Table 6). For the most-traded fifth of companies, the median gap was 3.7 basis points before the change and 8.0 after; on the first day of the auction it jumped to 14.3, and over the last ten days of the sample it averaged 8.3. For all other companies the median gap moved only from 13.1 to 13.8. A difference-in-differences regression on the daily medians gives an increase of 3.71 basis points relative to other companies (t = 11.1, standard errors robust to autocorrelation), and 3.67 (t = 12.6) when controlling for the size of each day’s market move.

Table 6. Gaps before and after the Closing Auction Session began on 2026-08-03.

	Before	After
Median absolute gap, the most-traded fifth of companies (bps)	3.7	8.0
Median absolute gap, all other companies (bps)	13.1	13.8
Mean signed gap, the most-traded fifth of companies (bps)	−0.4	−2.7
Identical closes on both exchanges, the most-traded fifth of companies	5.8%	5.4%

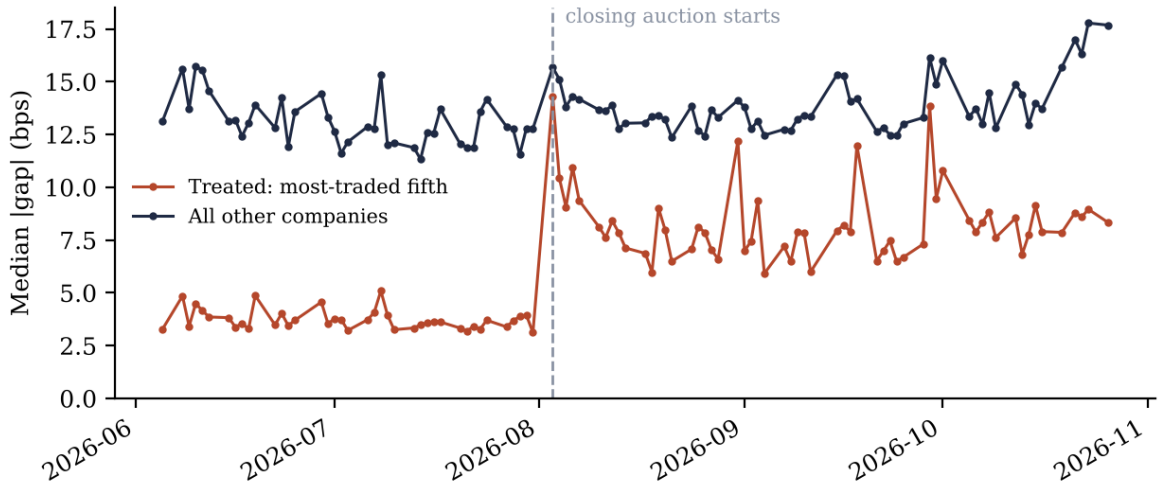


Figure 4. Median absolute gap each day for the most-traded fifth of companies and for all other companies, around the start of the Closing Auction Session (dashed line).

A plausible explanation is mechanical. A 30-minute volume-weighted average smooths out individual trades, and when most trading happens on NSE, both exchanges' averages track the same underlying price path. A closing auction instead sets the close from a single crossing of orders on each exchange, and each exchange runs its own auction with its own order book. Two separate auctions, one of them much smaller, can easily clear at slightly different prices. The data show one sign of this. An auction clears at the price of a limit order, and limit orders cluster on round numbers. For the most-traded fifth of companies, the share of closes at a whole rupee rose from 11.8% to 25.8% on NSE and from 4.7% to 23.9% on BSE, while for other companies it stayed near 5.6% and 7.1%. When each exchange's auction lands on its own round number, the two closes differ. If this is right, the closing auction made each exchange's close a better reflection of its own end-of-day demand, but made the two official closes of the same stock less consistent with each other. This matters for anyone who uses BSE closing prices for the most actively traded Indian stocks, including index providers, fund valuations and settlement prices.

9. What explains the puzzle?

Putting the tests together: BSE closes for thin stocks are often stale and lag the day's move on NSE; BSE's close, not NSE's, is the one that corrects the next day; the bias is largest on days with almost no BSE trading; and thin stocks trade on BSE mostly on days they fall on NSE, when a lagging BSE close sits above NSE's. Staleness explains why the gap moves with the market, while the remaining bias on quiet days is more consistent with how and when BSE trading happens for thin stocks. Intraday data would allow a direct test.

The practical lesson is the same as in the earlier paper. For a thinly traded dual-listed stock, BSE's official close is a noisy measure of the stock's value, and anyone using it, for example

to value a portfolio or to compute returns, should prefer NSE's close or check both. The closing-auction experiment adds a second lesson: for the most-traded stocks, the two exchanges' closes have diverged since the auction began, so the choice of exchange now matters for them too.

10. Limitations

The tests use daily closing data only, so they cannot see the trades behind each close. Identifying a thin BSE day from daily high, low, open and trade counts is approximate. The selection test compares days within a company, but days with and without BSE trading may differ in ways that are not captured by NSE returns and volume. The sample covers about one year, and the level shift means it may span two different regimes. The description of each exchange's closing-price rule follows the exchanges' published methodologies, and readers should consult the current rules directly. In the closing-auction experiment, the treated group is approximated by the most-traded fifth of companies rather than the official list of stocks with derivatives, and the period after the change is short.

11. Conclusion

The earlier paper found that India's two exchanges agree closely on the closing price of the same company, except that BSE tends to close slightly higher for thinly traded stocks. This paper tested four explanations using the same daily data. BSE closes for thin stocks are often stale and lag the day's move on NSE; BSE's close, not NSE's, is the one that corrects the next day; the bias is largest on days with almost no BSE trading; and thin stocks trade on BSE mostly on days they fall on NSE, when a lagging BSE close sits above NSE's. The introduction of a closing auction in 2026 also offered a natural experiment: for the most-traded fifth of companies, separate auctions on each exchange raised the typical gap between the two official closes from 3.7 to 8.0 basis points. The remaining open questions are best answered with intraday data.

Data and code

The data and the code that produces every number, table and figure in this paper are in the research/bse_puzzle folder of github.com/udit20112007/WORLD-CLOSE. Every number in the text is read from *results_puzzle.json*, so the paper always matches the data it was computed from.

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